

Basic Principles of Medicine II

Nervous System and Behavioral Sciences

IMCQ 8

St. George's University

1. A 26-year-old man is brought to the emergency department 1 hour after he sustained penetrating trauma to the back. Neurologic examination shows a loss of touch, vibration, and proprioception in his left leg. Which of the following structures is most likely damaged in this patient?

- A. Trigeminal lemniscus
- B. Anterolateral system
- C. Fasciculus gracilis
- D. Medial Longitudinal fasciculus
- E. Corona radiata

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2. A 67-year-old woman is brought to the emergency department because of a 4-hour history of a suspected stroke. CT scan of the head shows a small hemorrhage overlapping with the left superior cerebellar peduncle. Which of the following fiber tracts is most likely damaged in this patient?

- A. Anterior spinocerebellar
- B. Olivocerebellar
- C. Pontocerebellar
- D. Posterior spinocerebellar
- E. Vestibulocerebellar

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3. A 59-year-old man is brought to the physician by his wife because of a 3-month history of a resting tremor in his hands, which is worse on the right hand. Neurologic examination shows rigidity, slow movements, and reduced facial expression. Which of the following additional findings is most likely in this patient?

- A. Ptosis
- B. Shuffling gait
- C. Dysarthria
- D. Ataxia
- E. Nystagmus

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4. A 72-year-old man comes to the physician because of a 1-month history of weakness in his right upper limb. Physical examination shows right upper limb monoparesis. Imaging studies show a cerebral tumor. Which of the following tracts is most likely affected in this patient?

- A. Rubrospinal
- B. Vestibulospinal
- C. Reticulospinal
- D. Corticobulbar
- E. Corticospinal

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5. A 56-year-old man comes to the physician because of the sudden onset of double vision. Visual field testing shows intact vision in all four quadrants of the visual fields of each eye. Direct and consensual pupillary light responses are bilaterally normal. During primary gaze, the right eye is abnormally adducted. Which of the following is most likely observed on attempted rightward horizontal gaze?

- A. A
- B. B
- C. C
- D. D
- E. E

A



B



C



D



E



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SOM.MK.I.BPM2.1.NB.4.NSCI.0235 Relate the clinical correlations of Trochlear Nerve Palsy, Oculomotor Nerve Palsy, Abducens Nerve Palsy, Diplopia, Internuclear Ophthalmoplegia, PPRF Lesion and One-and-a-half Syndrome to the underlying functional and anatomical principles

A



B



C



D



E



6. A 62-year-old man comes to the physician because of a 2-day history of difficulty seeing clearly. Imaging studies show an infarction affecting the left optic tract. Axons connecting which of the following paired structures are most likely disrupted in this patient?

- A. Temporal hemi-retina (left eye); right lateral geniculate
- B. Temporal hemi-retina (left eye); left lateral geniculate
- C. Left lateral geniculate; right occipital lobe
- D. Right lateral geniculate; right occipital lobe
- E. Left lateral geniculate; left occipital lobe
- F. Right lateral geniculate; left occipital lobe

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7. A 67-year-old woman is brought to the physician by her daughter because of a 3-hour history of dizziness. Neurologic examination shows right unilateral hearing loss and pathological nystagmus to the left. The muscles of facial expression are weak on the right. The gag reflex is normal. Which of the following arteries is most likely affected in this patient?

- A. Left PICA
- B. Right PICA
- C. Left SCA
- D. Right SCA
- E. Left AICA
- F. Right AICA

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8. A 31-year-old woman comes to the physician because of a 2-day history of numbness on the left side of her body. Neurologic examination shows loss of touch and pain sensations on the left face, upper extremity, trunk and lower extremity. Neuroimaging shows a vascular lesion of the midbrain. In which area of the midbrain is the lesion most likely located in this patient?

- A. Left crus cerebri
- B. Left tectum
- C. Left tegmentum
- D. Right crus cerebri
- E. Right tectum
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9. A 52-year-old woman comes to the physician because of a 3-day history of double vision, particularly when looking to the right. Physical examination shows her right eye can abduct, but the left eye fails to adduct when looking to the right. When looking to the left, both eyes move normally. Vertical eye movements and pupillary responses are intact. Which of the following is the most likely site of the lesion?

- A. Left abducens nucleus
- B. Right abducens nucleus
- C. Left medial longitudinal fasciculus
- D. Right medial longitudinal fasciculus
- E. Left oculomotor nerve

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10. A 44-year-old man comes to the physician because of a 2-hour history of vertigo, nausea, and vomiting. Neurologic examination shows a rightward nystagmus. A diagnosis of Ménière's disease is made. Which of the following structures is most likely lesioned in this patient?

- A. Right labyrinth
- B. Left labyrinth
- C. Right cochlea
- D. Left cochlea
- E. Right external acoustic meatus
- F. Left external acoustic meatus

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11. A 28-year-old woman comes to the emergency department because of a 1-month history of vomiting, abdominal pain, and persistent hiccups. Despite prior treatment with anti-ulcer and antiemetic medications, her condition persists. MRI of the brain is shown; the arrow indicates an inflammatory zone. Which of the following structures is most likely affected in this patient?

- A. Area postrema
- B. Subfornical organ
- C. OVLT
- D. Posterior pituitary
- E. Subcommissural organ



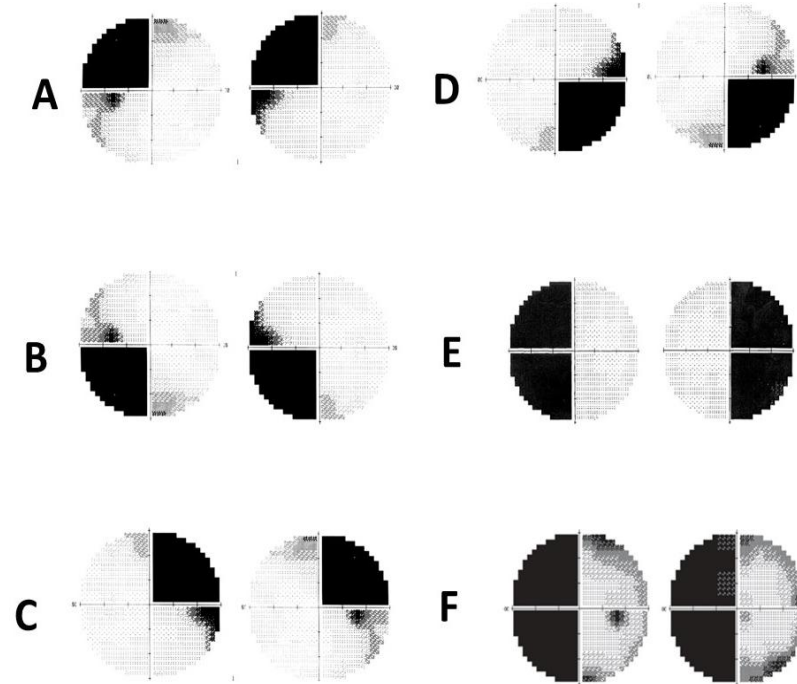
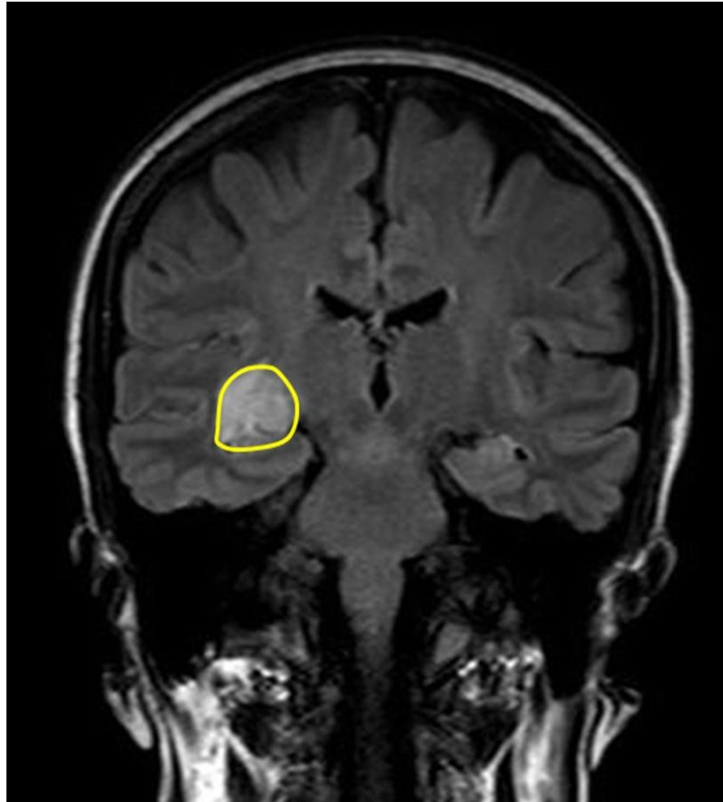
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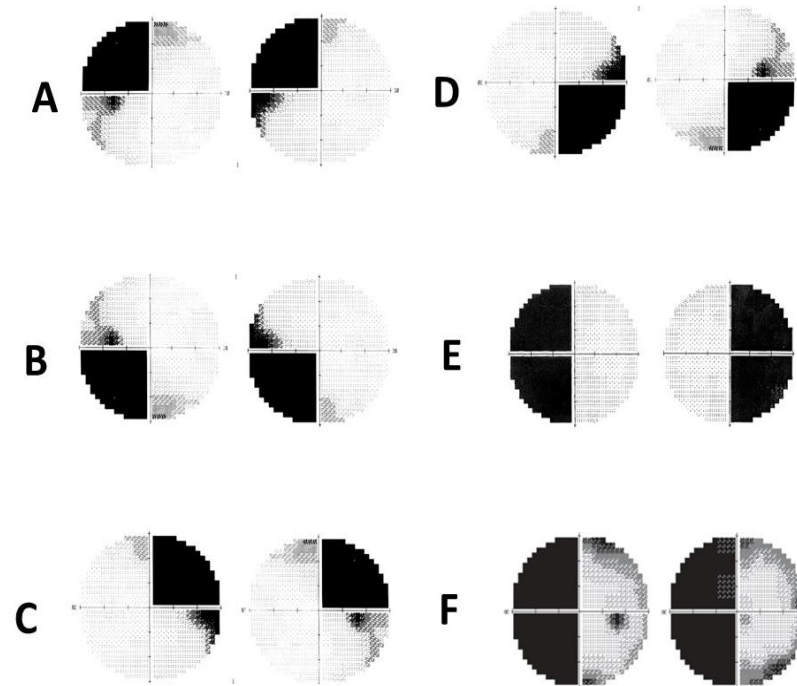
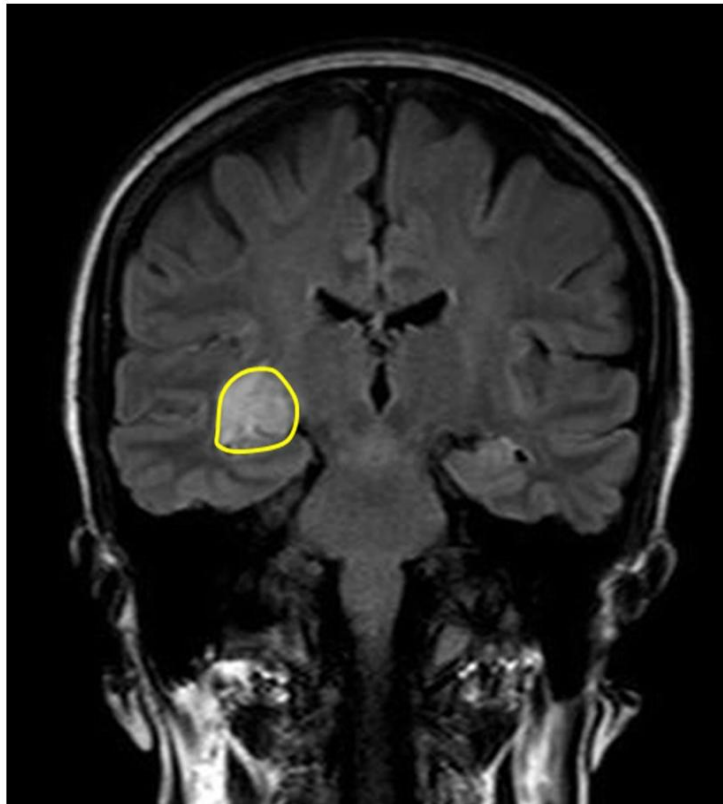
12. A 47-year-old woman comes to the physician because of a 3-week history of vision loss and headache. Neurologic examination shows a visual field deficit. MRI of the brain is shown; the yellow circle indicates a focal hyperintense lesion. Which of the following visual field maps shown best represents the visual fields in this patient?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F



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13. A 7-year-old boy is brought to the physician because of a 1-year history of blurred vision. Ophthalmic examination shows a defect in the focusing apparatus of the eyes. It is deduced that the focal point for light entering the eye falls anterior to the retina. Which of the following is the most likely diagnosis?

- A. Hyperopia
- B. Myopia
- C. Presbyopia
- D. Night blindness
- E. Color blindness
- F. Emmetropia

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14. A 17-year-old boy is brought to the emergency department 1 hour after he sustained a stab wound to the back. Neurologic examination shows muscle stretch reflexes are 2+ bilaterally in both the upper and lower extremities. Nociception is intact. Sensation to vibration is decreased in the left leg but intact in the right leg and also in both of his arms. Damage to which of the following labeled spinal cord structures shown in the images best explains the somatosensory deficits in this patient?

A. A

B. B

C. C

D. D

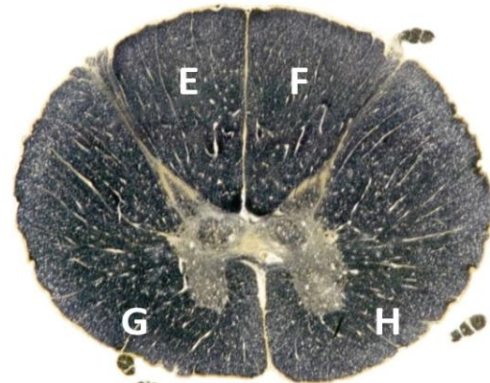
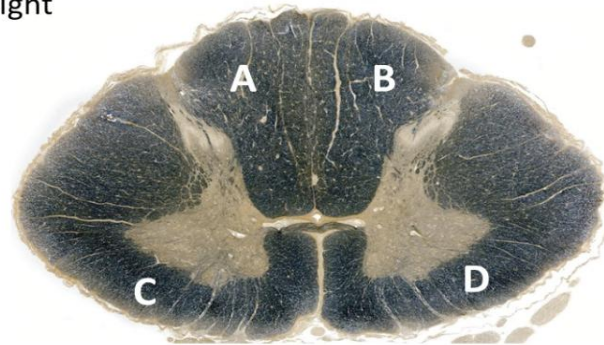
E. E

F. F

G. G

H. H

Right



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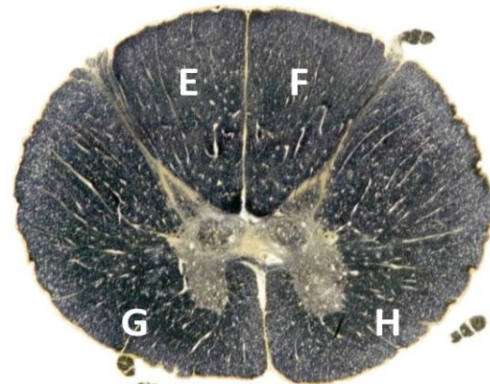
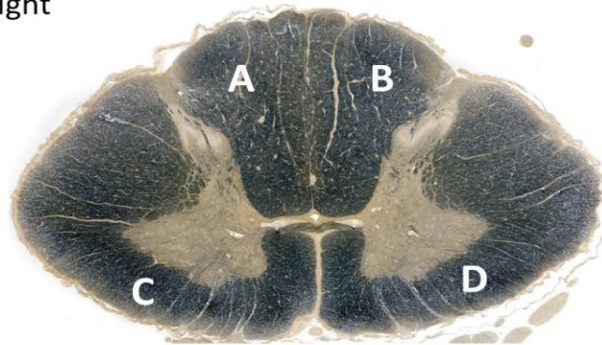
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G. G

H. H

Right



SOM.MK.I.BPM2.1.NB.1.NSCI.0128 describe the complete pathways for touch, vibration and proprioception, including the localization in different sections of the central nervous system